SAQ (3 cases):

CASE ABOUT ALZHEIMER.

Q1: diagnosis

Q2: list four differential diagnoses

Q3: Microscopic changes seen in Alzheimer

Q5: What grade of cognitive impairment in this case? Justify your answer

Q6: a question about episodic memory

Q7: a question about sementeric memory

MYOCLONIC AND TONIC-CLONIC EPILEPSY

Q1: Diagnose

Q2: Suggest two treatment

Q3: What is the difference between epilepsy and syncope

Q5: What is the difference between epilepsy and seizure attach

Q6: List two causes of epilepsy and mention the pathophysiological changes to one of them

Q7: what are the changes seen in EEG during epileptic attack

LEFT LATERAL MEDULLARY SYNDROME

Q1: diagnosis

Q2: Mention the reason for the following symptoms in this case.

a. Ptosis
b. Soft palate deviation
c. Loss of pain and temp sensation in the right side of the body
d. Increase pain sensation in the left side of the face
e. Paralyzed vocal cord

Q3: what is the artery involved in this case.

Q4: What are the changes that are seen in the MRI.

Case:

A 30 year-old man experienced weakness in his right leg that worsened over time till he couldn't move it. He also had absent feeling of position and vibration in the right leg till the level of the umbilicus and he had absent sensation of pain and temperature in the left leg. Also he had +ve Babinski sign.

Q1: which part of the nervous system was involved? Give 2 clues from the case that support your answer.

Q2: what is the tract that carries pain and temperature sensation?

Q3: explain briefly the pathway of pain transmission.

Q4: what is the tract that carries vibration and position sense?

Case:

A man had a stroke and he was unable to talk fluently, he had slurred slow speech and couldn't move his face and right hand, he had normal forehead furrow. He was able to follow commands like touch your head with your left hand.

Q1: what artery was involved?

Q2: what investigations would you do to confirm your diagnosis and why?

OSPE:

24 stations

1. Spinothalamic tract
2. Cerebral cortex histology
3. Purkinje is output from the cerebellar cortex
4. Peripheral nerve histology
5. Where is the lesion in Kluver Bucy syndrome, bilateral temporal lobe
6. A person has difficulty swallowing and talking, where is the lesion? 9th CN glossopharyngeal on the model
7. Where does this structure end "pointing to the spinal cord" and list 2 blood supplies to it
8. Injury to the arterial supply to the paracentral lobule will lead to? Which sinus drains directly to jagular veins → sigmoidal
9. Angiogram of cerebral blood flow and they asked us to point out to the artery responsible if the patient has aphasia and problem in moving arms
10. Which area of internal capsule contains corticonuclear fibers
11. Nausea and vomiting and paralyzed vocal cord and no swallowing → point out to vegus nerve in medulla
12. Physiology station about the tools used in measuring conduction velocity
13. Measure conduction velocity and why do we need 2 points for measuring it? (time : 7.4 msec and 2.4 msec , distance: 250 mm)
14. Myopathy shown a graph and give 2 reasons
15. A man is a pigeon breader and he is having high fever with meningitis? What is the diagnosis fungal meningitis. What is the causative agent genus and species cryptococcus neoformans.What is the basis of the stain? Capsular indian ink stain
16. Which cell act as macrophage in CNS
17. What nucleus of the cerebellum is responsible for correction of balance? Pointing to intracerebellar nuclei answer : fastigial nucleus
18. Which of the following give rise to the motor cells? Pointing to basal plate, alar plate, and the germinal neuroepithelial layer.
19. What is the clinical significance of the following structure pointing to the cavernous sinus? Infection in the paranasal sinus can be transmitted through the opthalamic veins to the intracranial fossa
20. The patient has difficulty abducting the eye where is the lesion? Abducent nerve on the brain stem model
21. Angiography showing basilar artery
22. Histology showing peripheral nerve and cerebral cortex and give 2 points of identification
23. Embryology pic: which part develops into sensory
24. Which part of the cerebellum is the major output?

MCQs 83 MCQs

Pathology:

1. A child with severe headache and fever. CSF showed increase WBCs (lymphocytes 94%) and protein and decrease glucose. What are the changes seen here? Caseas necrosis
2. Patient with ascending paralysis → GBS
3. Case on epidural hematoma
4. Case of meningioma
5. A patient with optic nerve problems and retina pigmentation. Where is the mutation? NF1 gene
6. Case about entrapment neuroma
7. Case about MS
8. Fungal meningitis what is the probable agent causing meningitis in a person taking corticosteroids → C. neuforms
9. Senario of vascular dementia
10. Chorea was seen in a patient what expected changes will be associated? Atrophy of the caudate nucleus

Anatomy:

1. A case with left temporal and parital lesion (Gestman syndrome(
2. Lession to which structre cause pure sensory loss in one side of the body → VPL
3. Hippocampus is connected to the mammillary bodies by? Fornix
4. Patient with loss of speech and muscle movement of face and arm. Which artery is affected? Middle cerebral artery
5. Which of the sensory structures derived from alar plate → Superior colliculus
6. Lacunar infarcts → pure sensory in the ventroposterior
7. Where do the motor cells come from? Basal plate
8. From where does the dorsal root ganglia arise? Neural crest cells
9. tumor in the interhemispheric fissure that affect cortex in both sides. What will be the symptoms?

Psychiatry:

1. If a patient starts crying in the psychiatric interview. What should you do? NOTHING
2. A patient doesn't answer questions probarly. When the dr asked him "when did you sleep last night" he said "I slept in a king size bed". What does he have? Tegmential thoughts
3. A patient is thinking that someone is trying to get him. What type of dellusion he has? Presecure
4. A patient with schizophrenia.
5. You diagnose psychiatric disorder based on what? Symptomatology

Biochem:

Which receptor can't be activated directly? NMDA receptor

Physiology:

1. Which of the following substances that if you add to extracellular fluid near a nerve causes rapid depolarization? Choises: Ca, K, Na or Cl
2. If you open post synaptic K channels, you will have → slow IPSP
3. Stage 2 of sleep
4. Delta waves
5. PTO area is responsible for ?
6. Bilateral loss of hippocampus leads to ? loss of long term potentiation
7. Lateralization mostly involves? Speech
8. What type of fibers transmits signals to arterial vessel wall? C fibers
9. Which area if it is stimulated, it cause motor movement in one joint? Premotor area
10. Patient can't repeat words; visual and auditory comprehension is affected. Which type of aphasia does this patient have?
11. Which reflex send inhibitory fibers?
12. Effector of the baroreceptor reflex pass through …..
13. Bradycardia and high blood pressure is seen in → increased intracranial pressure
14. You give a child a paper with numbers and one latter. She immediately picked up the letter. What type of attention is this? Object-based attention
15. Which nucleus of hypothalamus is responsible for maintaining sleep? Suprachiasmatic neucleus
16. Which of the following if it's damage, it leads to memory impairment → dorsal medial nucleus
17. The trick where the doctor presents an object and then hides it, based on what the patient would remember? Prefrontal cortex
18. Increase in extracellular Na does not affect the resting membrane potential because: membrane is not permeable to Na no matter how much it's increased

Pharma:

1. Sumatriptan binds to which receptor 5-HT1D and 5-HT1F
2. What would you give to a patient with panic attacks as an immediate treatment → citalopram
3. An opioid that causes psychometric symptoms → pentazocine
4. Dextamethazone is given with antibiotic treatment to reduce hearing loss in → H. influuenzea infection in pediatrics
5. Vanelfaxine binds to which receptor → serotonin and NE receptors
6. COMT metabolize which neurotransmitter → Dopamine
7. typical antipsychotic is binds to D2 receptors
8. Which of the following stimulants drugs has clinical indication → ketamine
9. Citalopram is → selective serotonin reuptake inhibitor

Clinical:

1. Patient with subarachnoid hemorrhage. How you investigate? Using CT without contrast

Micro:

Shingles is due to the reactivation of the virus: VZV (herpes zoster was an option to confuse b/c it's a condition not a virus!)

Radiology:

1. Which type of MRI you see CSF darkest → FLAIR
2. In subdural hemorrhage you see in CT → biconvex shape without contrast
3. The line in the posterior border of spinal cord of vertebral column is seen in CT as→ spinolaminar line